

CEREBROVASCULAR RESPONSES TO HYPERCAPNIA AND ACUTE AEROBIC AND RESISTANCE EXERCISE IN ADULTS LIVING WITH ATRIAL FIBRILLATION

Tori Roberts¹, Hannah J. Thomas², Lewan Parker³, Michelle A. Keske³, Kimberley L. Way³

¹ School of Exercise and Nutrition Sciences, Deakin University, Geelong VIC, Australia

² Centre for Healthy Ageing, Health Futures Institute, Murdoch University, Australia;

³ Institute for Physical Activity and Nutrition, Deakin University, Geelong VIC, Australia

Email: toroberts@deakin.edu.au

INTRODUCTION

Atrial fibrillation (AF) is the most common sustained cardiac arrhythmia globally. It has previously been demonstrated that individuals with AF have reduced cerebral blood flow (CBF) and cerebrovascular reactivity to carbon dioxide which are linked to an increased risk of cerebrovascular diseases such as stroke and vascular dementias. Current cerebrovascular disease management for this population includes anti-coagulant medication to address stroke risk. However, there remains limited treatment and prevention strategies addressing other cerebrovascular conditions adults with AF are at risk of. Exercise has well-established cardiovascular benefits that address risk factors associated with cerebrovascular disease such as reduced blood pressure and arterial stiffness and improved vascular function. As such, exercise may serve as an adjunct strategy to reduce the risk of cerebrovascular disease in the AF population. Therefore, characterising the functional cerebrovascular hemodynamic responses to acute aerobic and resistance exercise and mechanisms that assess cerebral vascular function (i.e., shear mediated vasodilation) are important to investigate in adults with AF. We aimed to i) characterise endothelium-dependent cerebrovascular function (cerebral flow-mediated dilation [cFMD]) to a transient hypercapnic stimulus in AF; and ii) assess the acute effect of aerobic and resistance exercise on middle cerebral artery velocity (MCAv) in adults living with AF.

METHODS

Adults with AF and controls were recruited. MCAv, blood pressure, heart rate and end-tidal carbon dioxide ($P_{ET}CO_2$) were collected continuously throughout each assessment. Transcranial Doppler was used to measure MCAv in the left and right middle cerebral artery. Vascular duplex ultrasound assessed right and left

internal carotid artery (ICA) blood flow during baseline and the cFMD assessment. First, ten minutes of resting baseline measurements were obtained in the supine position. Following this, participants' cFMD was assessed using a hypercapnic assessment where they breathed a gas mixture made up of 9% carbon dioxide (21% oxygen and balanced nitrogen) for 30 seconds. Participants then completed an acute bout of aerobic exercise (three minutes at 40%, 60% and 80% peak oxygen uptake) and resistance exercise (three sets of six repetitions at 40%, 60% and 80% one-repetition maximum) with a 45-minute washout period between exercise modalities.

RESULTS

Twenty-one participants (AF: $n=10$; controls: $n=11$) completed the study. Adults with AF had a significantly lower cFMD response ($5.10 \pm 3.23\%$) than controls ($10.36 \pm 1.62\%$, $p<0.01$) and longer left MCAv time to peak (AF: $45.43 \pm 19.61s$; control: $35.86 \pm 10.34s$, $p=0.01$). During aerobic exercise, adults with AF had no change in MCAv and mean arterial pressure ($p>0.05$) despite increases in heart rate and $P_{ET}CO_2$. Conversely, the control group had an increase in all variables in response to aerobic exercise (all $p<0.01$). During resistance exercise, MCAv and $P_{ET}CO_2$ remained unchanged despite increases in mean arterial pressure and heart rate ($p<0.01$) in both groups.

CONCLUSIONS

People with AF have impaired endothelium-dependant cerebrovascular vasodilation displayed by the lower cFMD response and longer time to peak. This serves as an independent risk factor for cerebrovascular diseases and may further explain the lack of change in bi-lateral MCAv during aerobic exercise. Preserved MCAv during resistance exercise may indicate stable autoregulation, suggesting resistance exercise may be safe for this population to perform.